Score	IB Grade

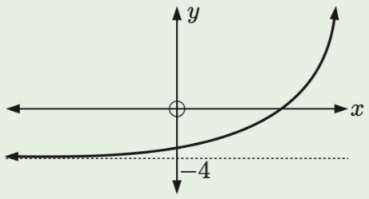
Name: ..... **Quiz 4** Mme Doux $11^{th}IB$

EXERCISE 1 [12 marks]

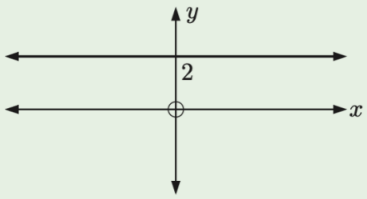
1 For each graph, state:

i the domain **ii** the range **iii** whether the graph shows a function.

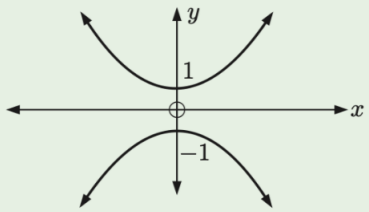
a



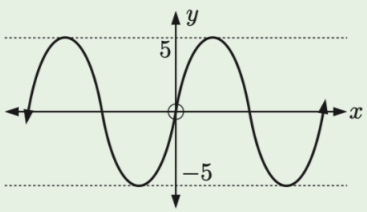
b



c



d



EXERCISE 2 **[4 marks]**

State the value(s) of x for which $f(x)$ is undefined:

(a) $f(x) = 10 + \dfrac{3}{2x-1}$

(b) $f(x) = \sqrt{x+7}$

EXERCISE 3 **[4 marks]**

If $F(x) = 2x^2 + 3x - 1$, find in simplest form:

(a) $F(-x)$

(b) $F(x+h)$

EXERCISE 4 [4 marks]

Given $f(x) = ax + \frac{b}{x}$, $f(1) = 1$, and $f(2) = 5$, find constants a and b.

EXERCISE 5 [6 marks]

Given $f(x) = \sqrt{1-x}$ and $g(x) = x^2$, find:

(a) $(f \circ g)(x)$ **[2]**

(b) the domain and range of $(f \circ g)(x)$ **[4]**

EXERCISE 6 [4 marks]

Given $f(x) = 2 + x$ and $g(x) = 3 - x$, find:

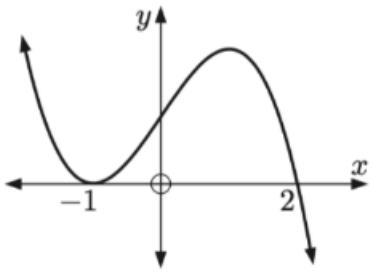
(a) $(f \circ f)(-5)$ **[2]**

(b) $(g \circ f)(3)$ **[2]**

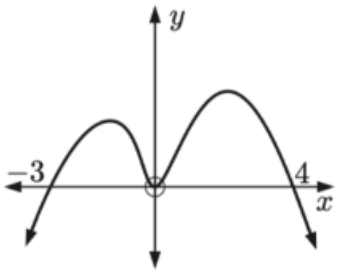
EXERCISE 7 [8 marks]

Draw sign diagrams for these graphs:

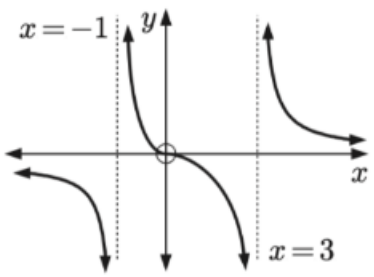
h



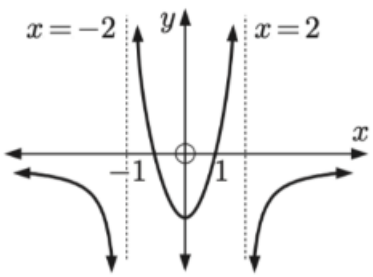
i



k



l



END !